

DevOps Engineer

Step 6: Build CI/CD Pipelines

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

CI/CD turns source changes into repeatable validation and delivery. A good pipeline reduces manual steps, gives fast feedback and provides an auditable path from commit to deployment.

AWS guidance recommends version control, testing/validation, configuration management, build/deployment systems, multiple environments and automation; it also emphasizes safe rollout strategies and monitoring.

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Core Concepts — Detailed Breakdown

- CI: automatically build/test changes when code is proposed or merged.
- CD: automate delivery/deployment after required checks.
- Pipeline stages: checkout → dependency install → lint/test → build → artifact/image → deploy → smoke test.
- Triggers: push, pull request, tag, schedule or manual approval.
- Secrets: inject credentials securely; do not commit them.
- Permissions: give CI tokens only the permissions needed.
- Artifacts: store build outputs or container images with traceable versions.
- Environments: separate staging and production; protect production.
- Rollback: know how to return to a known-good version.

What You Should Be Able To Do

- Create a workflow triggered by pull requests.
- Run tests and fail the workflow when tests fail.
- Build a Docker image.
- Publish an artifact.
- Deploy to staging.
- Add a protected production step.

Hands-on Project / Practice

- Create a GitHub Actions pipeline for a web application. Require tests before deployment, build an immutable image tagged with the commit SHA, deploy to staging, run a smoke test, then require approval for production.

Recommended References

- GitHub Actions Documentation — <https://docs.github.com/en/actions>
- GitHub Documentation — <https://docs.github.com/>
- AWS Well-Architected Framework — <https://docs.aws.amazon.com/wellarchitected/latest/framework/>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.